Review of "Memory Production under Coarse-Graining: A Balance Law for the Indivisibility Deficit"

1 Overall assessment

This is a strong and conceptually ambitious continuation of the preceding work on the indivisibility deficit and liftability profile. The paper addresses a natural question left open by the introduction of the dimension-matched charge: how does the measured failure of Chapman–Kolmogorov divisibility behave when the state space itself is coarse-grained? The principal conclusion—that neither monotone increase nor monotone decrease holds in general—is both important and convincingly motivated. More significantly, the manuscript replaces the failed monotonicity intuition with an exact balance law separating the coarse deficit into transmitted fine deficit and memory produced at the cut, while simultaneously decomposing the fine deficit into transmitted and severed components.

The central decomposition

$$\Delta V^{\lambda} = \Phi_{\lambda} + P_{\lambda}$$

is the main conceptual strength of the paper. Together with the orthogonal identity

$$\|\Delta V\|^2 = \|\Phi_{\lambda}\|^2 + \|\Xi_{\lambda}\|^2,$$

it gives a particularly clear interpretation of what coarse-graining can do to non-divisibility: existing deficit can be transmitted or severed, while additional deficit can be generated by unresolved dynamics at the cut. This resolves the apparent conflict between the intuition that eliminating variables generates memory and the competing intuition that coarse-graining destroys information.

The manuscript is also unusually effective in demonstrating that the hypotheses of its positive monotonicity result are genuinely necessary. The examples distinguish pure production, pure severing, dynamically closed reduction, and destructive interference, and the perfect-interference construction is especially useful because it shows that preservation of the transmitted component alone does not guarantee monotonicity. The one-parameter "interference dial"

further clarifies the geometry by exhibiting creation, conservation, annihilation, and subsequent growth within a single family.

There are, however, several places where the conceptual claims should be stated more carefully. In particular, the identification of P_λ with a "projection-free Mori–Zwanzig memory kernel" is stronger than the algebraic balance law itself and deserves especially precise qualification. The construction still uses the law-induced conditional-expectation projection $\Pi_r = R_r L$; what is projection-independent is principally the observed deficit and charge, together with the fact that this distinguished law projection reproduces the deficit for flat fine dynamics. The manuscript does partly acknowledge this distinction in Theorem 5.2, but some of the surrounding language risks suggesting a stronger independence from projection choice than has actually been established.

A second issue concerns the relation between the exact deficit-level results and statements about the charge h_n. The balance and Pythagorean identities are exact for the deficit cochain, whereas the transfer to the charge proceeds through two-sided comparison inequalities with dimension-, grid-, law-, and metric-dependent constants. The manuscript generally distinguishes these levels correctly, but several interpretive statements could more consistently emphasize that the exact conservation/balance structure belongs to ΔV, while the charge inherits quantitative control rather than the same literal additive identity.

The treatment of weak lumpability also deserves careful wording. The zero-production criterion exactly characterizes Chapman–Kolmogorov composition of the coarse two-time family when the fine family is flat. This is an important result, but the manuscript itself notes that pairwise Chapman–Kolmogorov consistency does not imply that the coarse process is Markov. Accordingly, describing the zero set simply as "weak lumpability" risks conflating a two-time conditional criterion with the full process-level notion unless the qualification "in its two-time formulation" is maintained consistently.

Overall, the mathematical architecture is coherent and the exact finite-state constructions substantially strengthen the paper. The main revisions I would recommend are therefore not changes to the central balance-law argument, but sharper delimitation of several interpretive claims: the precise sense in which the Mori–Zwanzig representation is projection-free, the distinction between exact deficit identities and charge-level comparison bounds, and the relationship between zero production, two-time Chapman–Kolmogorov composition, and conventional weak lumpability. Subject to clarification of these points and verification of the detailed constructions below, the paper makes a substantive contribution to the broader framework developed in the preceding manuscripts.

2 Major comments

The distinction between an exact balance law for the deficit and a balance law for the charge should be made sharper from the outset

The abstract and significance statement repeatedly describe the *charge* as obeying an exact balance law. However, the exact identities subsequently

introduced are identities for the deficit cochain:

$$\Delta V^{\lambda} = \Phi_{\lambda} + P_{\lambda}, \qquad \|\Delta V\|^2 = \|\Phi_{\lambda}\|^2 + \|\Xi_{\lambda}\|^2.$$

By contrast, the charge

$$h_n(V) = \operatorname{dist}(V, \mathrm{Flat}_n)$$

is related to the deficit norm through the two-sided Łojasiewicz-type comparison (7), rather than through an additive balance identity of its own.

This distinction matters because the constants and exponent in (7) depend on the dimension, grid, metric weights, and laws. Consequently, the exact conservation interpretation applies most directly to the deficit, while the charge inherits quantitative upper and lower control. I recommend modifying the abstract/significance statement so that it says explicitly that the deficit satisfies an exact balance law and that corresponding control of the charge follows through (7). This would prevent the reader from initially expecting an identity directly of the form $h_n(V^{\lambda}) = \cdots$.

The phrase "production is non-negative" is potentially misleading for a matrix-valued quantity. In the Introduction the manuscript states that the production term is "non-negative in the only sense a matrix-valued source can be, through its metric size." Since P_{λ} is a kernel difference with vanishing column sums, it has no natural entrywise positivity property, and later sections explicitly allow it to point opposite to the transmitted deficit.

It would therefore be cleaner to avoid describing production itself as non-negative. What is non-negative is simply its norm (or squared norm), as for every vector in a normed space. This does not by itself give P_{λ} the directional sign normally associated with a positive production term in a balance equation. The later destructive-interference example makes this especially important. A formulation such as "the magnitude of production is measured by $\|P_{\lambda}\|$" would be mathematically less suggestive.

The status of the law-weighted metric should be emphasized before using it to motivate the geometry of the cut. Equation (6) introduces the weights

$$\frac{p_i(u)}{p_j(v)}$$

and the manuscript correctly explains that this choice makes compression a conditional expectation and hence gives the orthogonal decomposition in Lemma 2.3. This is an elegant construction, but it is also load-bearing: Remark 2.4 explicitly states that the conditional-expectation and Pythagorean structure fails for generic admissible entry weights.

The paper should therefore distinguish more explicitly between metric-independent qualitative statements and metric-dependent geometric statements. The zero set of the charge/deficit is independent of the admissible metric, but orthogonality, the exact Pythagorean decomposition, the numerical size of flux and severing, and the contraction statement in the form used here depend specifically on the law-induced metric.

This is particularly relevant because the balance law is advertised as an intrinsic bookkeeping principle. The algebraic decomposition may be intrinsic to the chosen cut and family, whereas the interpretation of severing as an *orthogonal* conservation law is tied to the particular metric (6). I suggest making that distinction explicit near the first introduction of (5), rather than leaving the qualification mainly to Remark 2.4.

Lemma 2.3 would benefit from a clearer statement of the spaces between which C_i is acting. The conditional-expectation interpretation is central to the paper, but the presentation moves quickly between kernel differences, their density representatives f_A, the coarse kernel space, and the B_λ-measurable subspace of the fine L^2 space.

It would help to state explicitly that C_i itself is not literally the standard conditional-expectation operator on one fixed L^2 space. Rather, after identifying fine and coarse kernel differences with the corresponding density functions and using the block lift ι_i, compression corresponds to conditional expectation onto the B_λ-measurable subspace. This appears to be what the proof establishes, but making the identifications explicit would prevent the phrase "compression is conditional expectation" from being interpreted too literally.

The same clarification would strengthen the claim that C_i is a "metric surjection": the reader should be told precisely which Hilbert spaces and identifications are involved.

The terminology "compatible with the law" seems weaker than the phrase suggests. Definition 2.1 says that a cut is compatible with the law if all p_i are strictly positive, after which the block masses are automatically positive. But strict positivity is a property of the laws rather than a substantive compatibility condition between the cut λ and those laws. Under the standing assumption $p_i > 0$, every surjective cut is therefore compatible.

It may be clearer either to say directly that all cuts considered are surjective and the laws have full support, or to reserve "compatible" for a condition that actually relates λ to p. As currently phrased, a reader may initially expect compatibility to encode some dynamical or lumpability requirement, which it does not.

Example 3.2 establishes failure of monotonicity cleanly, but the role of strong lumpability should not be overinterpreted. The example correctly observes that the chain is not strongly lumpable because states 1 and 2, which belong to the same coarse block, have different transition probabilities into state 3. The subsequent explicit calculation then establishes that the coarse two-time family has a nonzero Chapman–Kolmogorov deficit.

It would be worth making clear that the failure of strong lumpability alone is not what proves that the particular observed family must have positive charge. Strong lumpability is sufficient for Markovianity under all initial distributions, but its failure does not by itself imply that a given initial law produces a non-flat coarse two-time family. Here the conclusion follows from the explicit nonzero deficit in (9), which is the stronger and appropriate argument.

A small wording adjustment would prevent readers from interpreting "not strongly lumpable" as logically equivalent to production of charge for the spec-

ified initial law.

The statement that the charge is "the metric size of its Chapman–Kolmogorov failure" could be qualified. The manuscript repeatedly uses this useful interpretation, but strictly

$$h_n(V) = \mathrm{dist}(V, \mathrm{Flat}_n)$$

is the distance of the *family* from the flat locus, whereas $\|\Delta V\|$ is the direct metric size of the Chapman–Kolmogorov residual. Equation (7) establishes two-sided control between these quantities but does not identify them.

I suggest consistently describing h_n as a metric measure of, or distance-based measure of, Chapman–Kolmogorov failure rather than its literal "metric size." This becomes particularly important in this paper because the exact balance identities are written in terms of ΔV, not $h_n(V)$, and maintaining the distinction from the beginning will make the later charge-level conclusions much cleaner.

The interpretation of Ξ_λ as deficit that "no coarse observable can see" should be stated relative to the chosen cut and metric structure. Theorem 4.2 gives the clean orthogonal decomposition

$$\|\Delta V\|^2 = \|\Phi_\lambda\|^2 + \|\Xi_\lambda\|^2,$$

and Remark 4.3 interprets Ξ_λ as the part of the fine deficit living on directions that no coarse observable can see. This is intuitive, but the wording is somewhat stronger than the construction itself.

More precisely, Ξ_λ is the component orthogonal to the block-measurable subspace associated with the particular cut λ, under the law-induced metric. It is therefore invisible to observables measurable with respect to that coarse partition, rather than intrinsically invisible to every possible coarse observation.

I suggest qualifying the language accordingly. This would also make the dependence of "severing" on the chosen coarse description explicit and align the interpretation with Lemma 2.3.

Theorem 4.5 uses two different notions of dynamical closure that should be distinguished terminologically. Definition 4.4 defines a cut as dynamically closed at level r when

$$LV(t \leftarrow r)Q_r = 0$$

on the entire fine state space. The converse part of Theorem 4.5 then proves that $P_\lambda \equiv 0$ only requires this operator to vanish on the accessible subspace A_r.

Thus zero production is not, strictly speaking, equivalent to "dynamical closure" as defined immediately beforehand; it is equivalent to dynamical closure restricted to the directions actually reached from earlier coarse states. The manuscript does state this qualification in the theorem, but the subsequent prose repeatedly describes production as vanishing "exactly when the cut is dynamically closed."

I recommend introducing separate terminology—for example, *strong dynamical closure* for Definition 4.4 and *accessible dynamical closure* for the

condition in Theorem 4.5—or modifying Definition 4.4 so that the accessible version is itself the operative definition. Otherwise the advertised "exact zero set" is slightly different from the formally defined property.

The identification of the zero-production set with weak lumpability needs to retain the "two-time formulation" qualification throughout. Corollary 4.6 is mathematically careful: for a flat fine family,

$$P_\lambda \equiv 0 \quad \Longleftrightarrow \quad V^\lambda \text{ is flat} \quad \Longleftrightarrow \quad \text{the two-time conditionals satisfy Chapman-Kolmogorov.}$$

The manuscript also explicitly notes that this does **not** imply that the coarse process Y is Markov.

This distinction is important enough that I recommend avoiding unqualified statements elsewhere that "zero production recovers weak lumpability." Conventional weak lumpability is normally a process-level Markov property for a specified initial distribution, whereas the result proved here characterizes composition of the two-time conditional family. The phrase "weak lumpability in its two-time formulation" is appropriate and should be retained consistently in the abstract, significance statement, Introduction, and conclusions.

Corollary 4.7 reinforces why the phrase "charge-level balance" may overstate what is proved. The result labelled "Charge-level balance" gives

$$\frac{1}{L_*}\big(\|P_\lambda\| - \|\Delta V\|\big)_+ \le h_n(V^\lambda) \le C_*(\|\Phi_\lambda\| + \|P_\lambda\|)^\theta,$$

with a cleaner two-sided estimate when the fine family is flat.

These are useful consequences of the exact deficit balance, but they are inequalities rather than a balance identity. In particular, there is no decomposition of the charge itself into transmitted, severed, and produced contributions.

I suggest renaming the corollary something such as "Charge-level bounds from the balance law" or "Charge-level consequences." This would make the logical hierarchy particularly clear: Theorem 4.2 provides the exact balance law; Corollary 4.7 translates it into quantitative control of the distance-to-flatness invariant.

Proposition 5.1 should distinguish carefully between an exact discrete expansion and identification with the full generalized Langevin equation. The expansion in (10) is an exact and useful decomposition obtained by inserting $\Pi_k + Q_k$ at the intermediate times. Calling it a "discrete generalized Langevin expansion" is reasonable as an analogy, but the manuscript then describes it as the finite-state analog of the Nakajima–Zwanzig equation and refers to the "full generalized Langevin structure."

It would help to specify exactly which structural features are being claimed to correspond. The classical Mori–Zwanzig/Nakajima–Zwanzig formalism normally involves projected evolution, orthogonal dynamics, and a temporally nonlocal memory term. Here the objects are finite-grid two-time conditional kernels and the expansion is an algebraic sum over resolved/fluctuation words.

The analogy appears meaningful, but the paper should distinguish an exact algebraic identity within the present framework from an equivalence to the usual

continuous-time generalized Langevin equation. A brief paragraph identifying the precise correspondence—and its limitations—would considerably strengthen this section.

The "projection-free Mori–Zwanzig kernel" terminology remains stronger than Theorem 5.2 actually establishes. Theorem 5.2 makes an interesting and precise statement: ΔV^λ is intrinsically defined from the observed family; the particular law-induced projection $\Pi_r = R_r L$ produces a memory kernel equal to that deficit; and other admissible sections generally give different kernels.

This does not mean that the memory-kernel construction itself is independent of projection choice. Rather, the observed deficit is projection-free, and there exists a distinguished projection—determined by the process law—for which the associated kernel coincides with it. Indeed, part (iii) explicitly demonstrates that changing the admissible projection changes the kernel.

I therefore recommend moderating phrases such as "the present setting dissolves the dependence" and "projection-free Mori–Zwanzig kernel." A more precise claim would be that the framework identifies a canonical, observation-defined representative of the memory effect and shows that the law-induced conditional-expectation projection realizes that representative exactly.

This distinction would preserve the genuinely interesting result without appearing to eliminate the standard projection dependence more generally than the theorem warrants.

The uniqueness statement in Theorem 5.2 should specify carefully the class over which uniqueness is claimed. The theorem concludes that the law projection is the unique admissible choice whose resolved model agrees with the observed conditionals, with the time-r law held fixed. The proof constructs detecting flat families when $\widetilde{R}_r \neq R_r$.

Because the uniqueness depends on the definition of an "admissible section"—non-negative columns supported on the blocks and satisfying $L\widetilde{R}_r = I_n$—and on varying over flat families while fixing the time-r law, these qualifications should accompany the uniqueness claim whenever it is summarized.

Otherwise "unique projection" may be read as uniqueness among arbitrary Mori, Zwanzig, Wiener, or other projection operators, which is much broader than what has been proved here.

Theorem 6.2 establishes monotonicity of the deficit norm, not literal monotonicity of the charge. Under hypotheses (S) and (C), Theorem 6.2 proves

$$\|\Delta V^\lambda\|^2 \geq \|\Delta V\|^2 + \|P_\lambda\|^2,$$

so monotonicity of the deficit norm is exact. The subsequent charge statement, however, again passes through (7), with different constants and potentially different exponents at successive dimensions.

Accordingly, the displayed lower bound for $h_{n_{k+1}}(V^{(k+1)})$ does not generally imply

$$h_{n_{k+1}}(V^{(k+1)}) \geq h_{n_k}(V^{(k)}).$$

What transfers exactly is preservation of *positivity* of the charge when the incoming deficit is nonzero, together with quantitative comparison through the dimension-dependent constants.

The manuscript should therefore be cautious in describing Theorem 6.2 as "monotonicity" of the charge. It is an exact monotonicity theorem for the deficit norm and a controlled propagation result for the charge. This is particularly important because the motivating question of the paper is explicitly phrased in terms of charge monotonicity.

"The deficit norm can grow only through production" should be interpreted locally rather than as a general causal statement.

Following Theorem 6.2, the manuscript states that the deficit norm can grow only through production. Within a fixed cut this follows naturally from

$$\Delta V^{\lambda} = \Phi_{\lambda} + P_{\lambda}$$

and the contractivity of the flux. However, because production can interfere either constructively or destructively with the flux, its mere presence does not imply growth; the coherence hypothesis is essential.

I suggest wording this as: *any increase beyond the transmitted fine deficit must be supplied by the production term*. This is slightly more precise and avoids suggesting that nonzero production necessarily increases the norm.

In Proposition 6.5, "no autonomous factor is discarded" does not appear to follow merely from failure of dynamical closure. The proposition states that nothing is severed and that "no autonomous factor is discarded, since the cut is not dynamically closed." Failure of dynamical closure certainly shows that the particular cut is not of the dynamically closed type occurring in Proposition 6.4. However, it is not immediately obvious that this alone proves the stronger structural statement that no autonomous factor is being discarded under every reasonable meaning of autonomous factor.

If "discarding an autonomous factor" is defined specifically as the product projection of Proposition 6.4, then the statement is fine but should be phrased in those terms. Otherwise an additional argument seems necessary. I recommend clarifying the definition being used here, since the point of Proposition 6.5 is precisely to isolate destructive interference from severing/autonomy as an alternative mechanism.

The inference from nonzero production to "no autonomous factor is discarded" should be stated more carefully. In the completion of Proposition 6.5, the manuscript calculates $(P_{\lambda} \neq 0)$,concludes that the cut is not dynamically closed, and then states that, by Proposition 6.4(i), the cut therefore discards no autonomous factor.

Proposition 6.4 establishes that projection away from an independent autonomous factor produces a dynamically closed cut and hence zero production. Its contrapositive excludes the particular product-projection mechanism described in Proposition 6.4 when $(P_{\lambda} \neq 0)$. It does not obviously exclude every broader interpretation of an "autonomous factor."

I therefore suggest replacing "the cut discards no autonomous factor" with

the more precise statement that the cut is not an autonomous-factor projection of the type characterized in Proposition 6.4. This is sufficient for the intended sharpness argument without making a stronger structural claim than the proposition establishes.

The interpretation of the interference dial as traversing the "entire phase diagram" is somewhat stronger than the calculation establishes. Remark 6.6 is a useful addition because the parameter (x) continuously changes the relative orientation and size of transmitted deficit and production. It exhibits production from a flat fine family, norm conservation, perfect cancellation, and subsequent growth with reversed orientation.

However, the phrase "one family walks the entire phase diagram of the balance law" should perhaps be softened. The construction explores a one-dimensional slice in which

$$P_\lambda = -\frac{1}{64}\delta_0, \qquad \Phi_\lambda = x\delta_0,$$

so production and flux remain collinear throughout. The general balance law permits arbitrary relative orientations and magnitudes of Φ_λ and P_λ, as well as nonzero severing.

The example therefore traverses the complete range from constructive to destructive interference **within this collinear family**, rather than the entire geometric phase space of the balance law.

The discussion of exact arithmetic should distinguish exact verification of the displayed examples from verification of the general theoretical results. Section 7 emphasizes that every quantity is computed using exact rational arithmetic and that the identities can be checked directly from the displayed matrices. This is a strength of the examples and makes the counterexamples particularly transparent.

The wording should nevertheless avoid suggesting that exact rational verification of the examples independently verifies the general balance law or the associated structural theorems. The table confirms that the four examples instantiate the claimed regimes; the general results still depend on the preceding algebraic proofs.

A slight rephrasing would make clear that the table provides exact *illustrations and consistency checks* of Theorem 4.2 rather than an additional verification of the theorem itself.

The four rows of Table 1 are useful canonical examples, but "four pure regimes" should be defined more precisely. The table labels production, severing, closure, and interference as the four pure regimes of the balance law. This is intuitively effective, but mathematically the general decomposition involves continuous values of

$$|\Phi_\lambda|, \qquad |\Xi_\lambda|, \qquad |P_\lambda|, \qquad \langle \Phi_\lambda, P_\lambda \rangle,$$

and therefore does not naturally partition into exactly four exhaustive regimes.

It would be clearer to describe the rows as four *extremal or representative mechanisms* exhibited by the balance law: pure production, pure severing,

complete closure, and perfect destructive interference. This retains the usefulness of the table without implying that all possible behavior belongs to one of four discrete classes.

The proposed statistical test ($P_\lambda = 0$) should be described as a test of two-time closure rather than lumpability without qualification. The open-problems section proposes using estimates of (P_λ) to construct a practical test of "dynamical closure, hence of two-time weak lumpability." This is a promising direction, but the distinction developed earlier in the manuscript remains important.

A test based entirely on two-time transition families cannot in general establish the full process-level Markov property of the lumped process. What it can test is the Chapman–Kolmogorov consistency of the observed two-time family, or equivalently the zero-production condition under the hypotheses of Corollary 4.6.

The proposed statistical interpretation should therefore retain the term "two-time" prominently and perhaps explicitly state that higher-order conditional structure would be required to test genuine process-level Markovianity.

The claimed equivalence between minimizing production and minimizing observed charge is too strong. The final open problem states that, among cuts of fixed coarse dimension, minimizing production is "equivalently, by Corollary 4.7, the charge of the observed family when the fine family is flat."

Corollary 4.7 establishes only

$$\frac{1}{L_*}|P_\lambda| * * \leq * * h_n(V^\lambda) * * \leq * * C_*|P_\lambda|^\theta.$$

These inequalities establish equivalence of the **zero sets** and two-sided quantitative control, but they do not in general imply that two cuts are ordered identically by $(|P_\lambda|)$ and $(h_n(V^\lambda))$, especially because the constants and possibly the induced laws/metrics may vary with the cut.

Thus minimizing production is not obviously equivalent to minimizing charge as an optimization problem. The manuscript could instead say that production minimization is *closely related to* charge minimization, or that the two objectives have the same zero set for flat fine dynamics. A genuine equivalence of minimizers would require an additional argument.

The proposed continuous-grid "production rate" is appropriately identified as open, but the expected scaling should not be presupposed. The manuscript suggests that, upon grid refinement, production "should scale to a production rate" and the balance law to a continuity equation. This is an interesting direction, but the present finite-grid theory does not establish the scaling required for such a limit.

Depending on the regularity and short-time structure of the transition family, the deficit and production terms could potentially scale at different orders in the time increment. Before a nontrivial production rate can be defined, one would need to identify the appropriate normalization and establish convergence.

I suggest presenting this explicitly as the first mathematical problem in the continuous-grid extension rather than assuming that a rate necessarily exists.

This would make the open problem more precise and avoid building the proposed continuity equation into the conjecture.

The paper would benefit from a short concluding synthesis before the open problems. The manuscript moves directly from the exact computations and Table 1 into the list of open problems. Given that this paper appears to close an important conceptual loop in the broader sequence of manuscripts, a short conclusion would improve the presentation.

Such a conclusion could state clearly that the principal exact object is the deficit-level balance

$$\Delta V^{\lambda} = \Phi_{\lambda} + P_{\lambda},$$

that coarse-graining changes apparent memory through the three mechanisms of transmission, severing, and production/interference, and that the dimension-matched charge inherits quantitative information through its comparison with the deficit rather than through a literal additive conservation law.

This would also provide an appropriate place to reiterate the limitations established within the paper itself: two-time flatness need not imply process-level Markovianity, projection-freeness refers to the intrinsic observed deficit and its canonical law-induced realization, and the current theory concerns the dimension-matched charge rather than the full liftability profile.